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Chapter-1

Overview

- The provided Python script implements a **Cafe Management System** with a graphical user interface (GUI) using the Tkinter library. This system enables cafe staff to:
 1. Display a menu with item images, prices, and quantity selectors.
 2. Allow users to calculate bills based on selected items and
 3. Generate a receipt showing the order summary.
 4. Provide options to clear the form or finalize the order with a confirmation dialog.

- The program ensures an intuitive interface, incorporates essential features for managing cafe orders, and provides visual appeal through integrated item images.

1.1 Code Structure

1.1.1 Libraries Used

1. Tkinter:

Core library for creating GUI components like labels, buttons, entry fields, and frames.

Functions used: Tk, Label, Button, Frame, Entry, Text, etc.

2. Pillow:

Used for handling and resizing images with the Image and ImageTk modules.

3. MessageBox (from Tkinter):

Provides dialog boxes for displaying messages like errors or success notifications.

Chapter-2

Functionalities

2.1 Menu Display

The menu includes:

- Items listed with names, prices, and corresponding images.
- Quantity input fields for each item.
- Layout arranged in a grid format within a frame.

Each menu item is represented by:

- Layout arranged in a grid format within a frame.
- A label showing the name and price.
- An entry field to input the desired quantity.

2.2 Bill Calculation

The `calculate_bill()` function:

- Iterates through all menu items and their respective quantities.
- Calculates the total cost by multiplying item price by quantity.

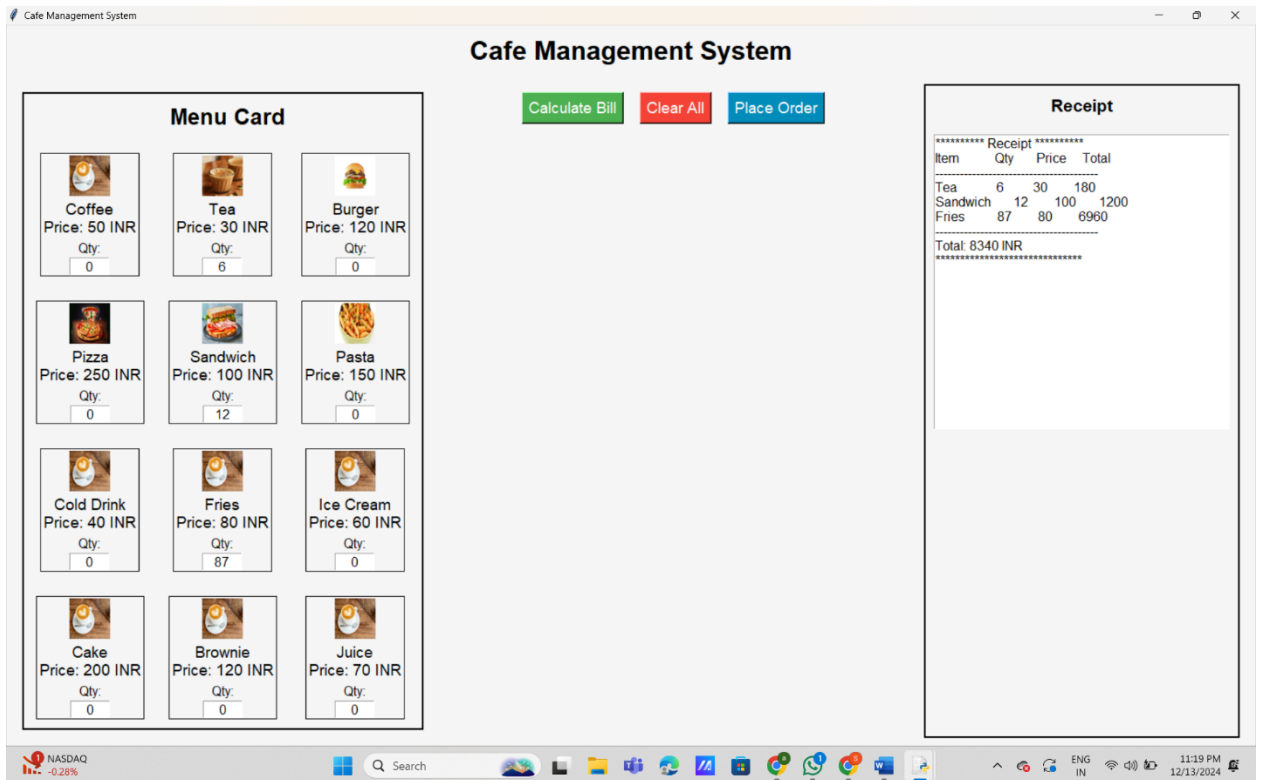
Displays a detailed receipt in a text widget, including:

- Item name, quantity, price per unit, and total cost for each item.
- The total amount payable.

2.3 Receipt Generation

The receipt is displayed dynamically in a Text widget and includes:

- A header ("Receipt").
- Itemized details.
- A final total amount.



2.4 Reset Functionality

The `clear_all()` function:

- Resets all input fields to 0.
- Clears the receipt display.

2.5 Order Placement

The `place_order()` function:

- Checks if any item quantity is greater than 0 before proceeding.
- If no items are selected, an error message is displayed.
- If valid, a success message is shown, and all fields are reset.

Chapter-3

GUI Layout

3.1 Main Window

- The main window is created using Tk() with a title, fixed dimensions (1000x600), and a light background color (#f5f5f5).

3.2 Frames

The layout uses Frame widgets to organize content:

- **Menu Frame:**
 - * Displays menu items in a grid with images, names, prices, and quantity fields.
- **Order & Receipt Frame:**
 - * Contains a text widget for showing the generated receipt.
- **Order & Receipt Frame:**
 - * Hosts the action buttons: "Calculate Bill," "Clear All," and "Place Order."

3.3 Widgets

- **Labels:**
 - For displaying static text like titles, menu items, and receipt sections.
- **Entry:**
 - Used for quantity input, bound to IntVar for seamless data binding.
- **Text:**
 - Multi-line widget for displaying the receipt.
- **Buttons:**
 - Trigger actions like bill calculation, clearing fields, and order placement.

Chapter-4

Key Features and Functionality

4.1 Menu Items

- **Dynamic Representation:** Items are loaded from the menu_items dictionary, where each item includes its price and image path.
- **Images:** Displayed for each item after resizing to a uniform size (50x50 pixels).

4.2 Bill Calculation

- Computes total cost dynamically based on user input.
- Receipt format includes:
 - Columns for item name, quantity, price, and total.
 - A total payable amount at the end.

4.3 Error Handling

- Prevents order placement when no items are selected.
- Ensures fields are cleared properly when "Clear All" is clicked.

Chapter-5

Strengths of the Code

5.1 Readability:

The code is well-structured and easy to follow.

Function names and variable names are descriptive.

5.2 Dynamic Design:

Menu items are generated dynamically from the menu_items dictionary, simplifying scalability.

Images, prices, and item names are seamlessly integrated.

5.3 User Feedback:

Error and success messages provide clear feedback to the user.

5.3 Reusability:

Centralized data (menu_items) ensures easy updates or additions to the menu.

Chapter-6

Areas for Improvement

6.1 Validation

Validate the input in quantity fields to ensure only numeric values are entered.

Non-

numeric or negative inputs can cause unexpected behavior.

6.2 Example:

python

Copy code

```
def validate_quantity(qty):  
    try:  
        return max(0, int(qty)) # Convert to int and ensure non-negative  
    except ValueError:  
        return 0
```

6.3 Layout Enhancements

Improve visual appeal by using ttk widgets for a modern look.

Add scrollbars to the receipt area for handling lengthy orders.

Save Receipt:

Provide functionality to save the generated receipt as a text or PDF file.

6.4 Example:

```
python  
  
Copy code  
  
def save_receipt():  
  
    with open("receipt.txt", "w") as file:  
  
        file.write(receipt_text.get(1.0, END))
```

6.4 Tax and Discounts

Add optional fields for taxes or promotional discounts.

6.6 Error Handling

Handle scenarios where image paths are incorrect or files are missing.



Chapter-7

Testing Plan

7.1 Test Cases

Test ID	Scenario	Expected Outcome
TC01	Valid input for all fields	Receipt displays correct calculations.
TC02	No quantity entered	Total shows 0; error on order placement.
TC03	Non-numeric quantity input	Input resets to 0 or shows an error message.
TC04	Clear All clicked	All fields reset; receipt cleared.
TC05	Images missing for menu items	Application runs but skips displaying images.

Conclusion

The Cafe Management System is a functional and visually appealing application designed for small to medium cafes. While it meets the basic requirements, integrating features like input validation, tax handling, and receipt saving can further enhance its utility and robustness.

Cafe Management Systems

➤ Latta 8 ➤ Espresso 3 ➤ Iced Latte 4 ➤ Vale Coffee 7 ➤ Cappuccino 5 ➤ African Coffee 1 ➤ American Coffee 2 ➤ Iced Cappuccino 3	➤ Coffee Cake 1 ➤ Red Velvet Cake 10 ➤ Black Forest Cake 8 ➤ Boston Cream Cake 9 ➤ Lagos Chocolate Cake 3 ➤ Kilburn Chocolate Cake 4 ➤ Carlton Hill Chocolate Cake 5 ➤ Queen's Park Chocolate Cake 2	Receipt <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td>Receipt Ref:</td> <td>BILL229616</td> <td>11/06/2016</td> </tr> <tr> <td>Name</td> <td></td> <td>Cost of Items</td> </tr> <tr><td>Latta</td><td></td><td>8</td></tr> <tr><td>Espresso</td><td></td><td>3</td></tr> <tr><td>Iced Latte</td><td></td><td>4</td></tr> <tr><td>Vale Coffee</td><td></td><td>7</td></tr> <tr><td>Cappuccino</td><td></td><td>5</td></tr> <tr><td>African Coffee</td><td></td><td>1</td></tr> <tr><td>American Coffee</td><td></td><td>2</td></tr> <tr><td>Iced Cappuccino</td><td></td><td>3</td></tr> <tr><td>Coffee Cake</td><td></td><td>1</td></tr> <tr><td>Red Velvet Cake</td><td></td><td>10</td></tr> <tr><td>Black Forest Cake</td><td></td><td>8</td></tr> <tr><td>Boston Cream Cake</td><td></td><td>9</td></tr> <tr><td>Lagos Chocolate Cake</td><td></td><td>3</td></tr> <tr><td>Kilburn Chocolate Cake</td><td></td><td>4</td></tr> <tr><td>Carlton Hill Chocolate Cake</td><td></td><td>5</td></tr> <tr><td>Queen's Park Chocolate Cake</td><td></td><td>2</td></tr> <tr> <td>Cost of Drinks:</td> <td>(£, 58.59)</td> <td>Tax Paid: (£, 20.54)</td> </tr> <tr> <td>Cost of Cakes:</td> <td>(£, 76.74)</td> <td>Sub Total: (£, 136.92)</td> </tr> <tr> <td>Service Charge:</td> <td>(£, 1.59)</td> <td>Total Cost: (£, 157.46)</td> </tr> </table>	Receipt Ref:	BILL229616	11/06/2016	Name		Cost of Items	Latta		8	Espresso		3	Iced Latte		4	Vale Coffee		7	Cappuccino		5	African Coffee		1	American Coffee		2	Iced Cappuccino		3	Coffee Cake		1	Red Velvet Cake		10	Black Forest Cake		8	Boston Cream Cake		9	Lagos Chocolate Cake		3	Kilburn Chocolate Cake		4	Carlton Hill Chocolate Cake		5	Queen's Park Chocolate Cake		2	Cost of Drinks:	(£, 58.59)	Tax Paid: (£, 20.54)	Cost of Cakes:	(£, 76.74)	Sub Total: (£, 136.92)	Service Charge:	(£, 1.59)	Total Cost: (£, 157.46)
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£ 58.59	£ 20.54	
Cost of Cakes	Sub Total	
£ 76.74	£ 136.92	
Service Charge	Total Cost	
£ 1.59	£ 157.46	

Total	Receipt	Reset	Exit
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